SAT Math · Week 8 Practice Set

24 questions · 35 minutes

Name _____ Class _____ Date _____

1. No calculator

$$\frac{2x-5}{3}+\frac{x+1}{2}=7$$

According to the equation above, what is the value of $7x-7$?

Student-produced response: enter your answer as a number.

2. No calculator

A chemist mixes a 20% acid solution with a 50% acid solution to make 30 liters of a 40% acid solution. How many liters of the 20% solution are used?

(A) 10
(B) 12
(C) 15
(D) 20

3. No calculator

The graph of $y=f(x)$ is a parabola with vertex $(3,-4)$ that passes through the point $(5,4)$. What is the value of $f(0)$?

(A) 5
(B) 14
(C) 18
(D) 32

4. No calculator

What is the sum of all the solutions of the equation $\sqrt{3x+10}=x+2$?

Student-produced response: enter your answer as a number.

5. No calculator

$f(x)=6\cdot 16^{x/4}$

Which of the following is equivalent to the function f above?

(A) $f(x)=6(2)^x$
(B) $f(x)=6(4)^x$
(C) $f(x)=6(16)^x$
(D) $f(x)=6(64)^x$

6. No calculator

A data set has a mean of 40 and a standard deviation of 6. Each value in the set is multiplied by 3, and then 5 is subtracted from the result. What is the standard deviation of the new data set?

Student-produced response: enter your answer as a number.

7. CALCULATOR

A tap fills a rectangular tank at a constant rate of 4.5 liters per minute. The inside of the tank measures 1.2 meters by 0.5 meter by 0.9 meter. How many hours does the tap take to fill the tank? (1 cubic meter = 1,000 liters.)

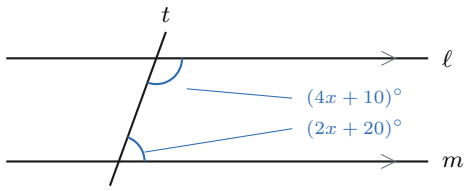
(A) 0.2
(B) 2
(C) 12
(D) 120

8. DESMOS

A ball is thrown upward from a platform. Its height, in feet, t seconds after it is thrown is $h(t) = -16t^2 + 48t + 64$. How many seconds pass between the moment the ball reaches its maximum height and the moment it hits the ground?

(A) 1.5
(B) 2
(C) 2.5
(D) 4

9. NO CALCULATOR



Note: figure not drawn to scale.

In the figure, lines ℓ and m are parallel and are cut by transversal t. What is the measure, in degrees, of the larger of the two marked angles?

(A) 25
(B) 30
(C) 70
(D) 110

10. NO CALCULATOR

In triangle ABC, point D lies on $\overline{AB}$ and point E lies on $\overline{AC}$, with $\overline{DE} \parallel \overline{BC}$. If $AD = 5$, $DB = 10$, and $DE = 8$, what is the length of $\overline{BC}$?

(A) 12
(B) 16
(C) 18
(D) 24

11. NO CALCULATOR

Triangle ABC is similar to triangle DEF, with $AB = 6$ corresponds to $DE = 15$. The area of triangle ABC is 28. What is the area of triangle DEF?

(A) 37
(B) 70
(C) 175
(D) 437.5

12. CALCULATOR

A surveyor whose eyes are 1.6 meters above level ground stands 40 meters from the base of a tower. The angle of elevation from the surveyor's eyes to the top of the tower is 35°. To the nearest tenth of a meter, what is the height of the tower above the ground?

(A) 20.6
(B) 24.5
(C) 28.0
(D) 29.6

13. NO CALCULATOR

In right triangle ABC, the right angle is at B, $AC = 26$, and $\sin A = \frac{5}{13}$. What is the area of triangle ABC?

Student-produced response: enter your answer as a number.

14. NO CALCULATOR

In triangle PQR, side $\overline{QR}$ is extended beyond R to point S. The measures of $\angle PRS$, $\angle P$, and $\angle Q$ are $(7x - 5)°$, $(2x + 10)°$, and $(3x + 5)°$, respectively. What is the measure, in degrees, of $\angle PRQ$?

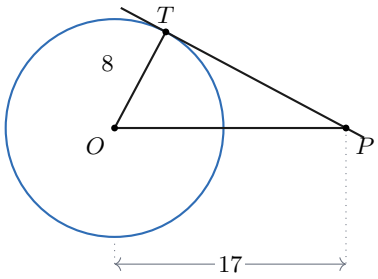
(A) 10
(B) 35
(C) 65
(D) 115

15. NO CALCULATOR

The equation $2x^2 + 2y^2 - 12x + 16y = 22$ represents a circle in the xy-plane. What is the radius of the circle?

(A) 6
(B) $\sqrt{47}$
(C) 12
(D) 36

16. NO CALCULATOR



Note: figure not drawn to scale.

In the figure, line PT is tangent to the circle with center O at point T. The radius of the circle is 8 and $OP = 17$. What is the area of triangle OTP?

(A) 60
(B) 68
(C) 120
(D) 136

17. NO CALCULATOR

A circle has diameter 24. A sector of the circle has an arc length of 10π. What is the area of the sector?

(A) 10π
(B) 60π
(C) 120π
(D) 240π

18. NO CALCULATOR

In the xy-plane, the points $(-1, 2)$ and $(7, 8)$ are the endpoints of a diameter of a circle. Which of the following is an equation of the circle?

(A) $(x - 3)^2 + (y - 5)^2 = 5$
(B) $(x + 3)^2 + (y + 5)^2 = 25$
(C) $(x - 3)^2 + (y - 5)^2 = 25$
(D) $(x - 3)^2 + (y - 5)^2 = 100$

19. No calculator

Points A, B, and C lie on a circle with center O. Inscribed angle $\angle ACB$ and central angle $\angle AOB$ both intercept arc AB. If $\angle ACB = (2x + 6)°$ and $\angle AOB = (7x - 12)°$, what is the measure, in degrees, of $\angle ACB$?

(A) 8
(B) 13.2
(C) 22
(D) 44

20. No calculator

A wheel rotates at a constant rate of $\frac{3\pi}{4}$ radians per second. How many complete rotations does the wheel make in 2 minutes?

Student-produced response: enter your answer as a number.

21. No calculator

A right circular cylinder with radius 12 centimeters contains water. A solid metal sphere with radius 6 centimeters is completely submerged in the water, and no water spills out. By how many centimeters does the water level rise?

(A) 1
(B) 1.5
(C) 2
(D) 8

22. No calculator

In the xy-plane, line k is perpendicular to the line $3x + 4y = 12$ and passes through the point $(6, 1)$. What is the y-intercept of line k?

(A) -7
(B) 5.5
(C) 7
(D) 9

23. No calculator

A circle is inscribed in a square. The area of the region inside the square but outside the circle is $36(4 - \pi)$. What is the perimeter of the square?

Student-produced response: enter your answer as a number.

24. No calculator

A right circular cylinder has a volume of 72 cubic centimeters. A second right circular cylinder has a radius 3 times the radius of the first and a height $\frac{1}{4}$ of the height of the first. What is the volume, in cubic centimeters, of the second cylinder?

(A) 54
(B) 162
(C) 216
(D) 648